

MOHAK AGARWAL

Software Engineer | Full Stack Developer

Noida, India | +91-9565814426 | agarwalmohak25@gmail.com | linkedin.com/in/mohak-agarwal-243474155

Professional Summary

Full Stack Engineer with **2.5+ years of experience** building scalable web applications using **Node.js, React.js, Next.js, TypeScript, and PostgreSQL**. Experienced in **REST APIs, Redis caching, backend optimization, scalable frontend architectures, and clean system design**. Familiar with **LLM workflows, GenAI tooling, RAG fundamentals, AI-assisted development, and workflow automation**.

Technical Skills

- **Languages & Frameworks:** JavaScript, TypeScript, Node.js, Express.js, React.js, Next.js
- **Frontend:** Tailwind CSS, Shadcn/ui, Zustand, TanStack Query, Zod, Progressive Web Apps (PWA)
- **Backend:** REST APIs, Redis Caching, Clean Architecture, Microservices, Keycloak, AWS S3
- **Databases:** MongoDB, PostgreSQL
- **AI & Modern Workflows:** GenAI Tooling, LLM Workflows, RAG Fundamentals, Prompt Engineering, AI-assisted Development
- **Tools & Practices:** Git, Agile/Scrum, System Design, Data Structures & Algorithms

Professional Experience

Software Engineer

April 2026 – Present

Jasper Colin

Noida, India

- Developed enterprise-grade full-stack modules using **Node.js, React.js**, and scalable API-driven architectures across Agile delivery cycles.
- Built reusable frontend and backend components, reducing feature implementation effort by approximately **30%**.
- Collaborated with cross-functional teams to translate business requirements into modular, production-ready systems.
- Improved engineering workflows using **AI-assisted development** practices and reusable architecture patterns.

Software Engineer

March 2024 – April 2026

Ksolves

Noida, India

- Optimized mission-critical **Express.js APIs**, reducing average server load by **20%** and improving database response times by **32%**.
- Built a high-performance **PWA** handling **2,000+ daily transactions** using Zustand and TanStack Query.
- Engineered a custom media recording utility using JS Blob APIs supporting **10,000+ monthly active users**.
- Contributed to reusable UI systems, scalable API integrations, and frontend architecture improvements across enterprise applications.
- Conducted **50+ code reviews** and mentored junior developers on scalable frontend and backend practices.

Technical Projects

Ride-Hailing Backend Prototype | *Node.js, Redis, Geospatial APIs, Clean Architecture*

- Developed scalable backend APIs for real-time ride booking, driver availability tracking, and nearby-driver discovery using **Redis GEO indexing**.
- Implemented modular service layers using **Clean Architecture**, improving maintainability and backend scalability.
- Optimized driver lookup operations to achieve **sub-150ms response times** while handling **500+ simulated location updates** daily.

AI-Powered Survey & Feedback Management System | *Next.js, Node.js, LLM Workflows, NLP Pipelines*

- Built a multimedia feedback platform supporting text, audio, and video submissions from **1,000+ daily active users**.
- Integrated lightweight **LLM-powered summarization workflows** to automate structured feedback analysis.
- Designed backend processing pipelines for metadata extraction, sentiment grouping, and automated report generation.

PG-Manager — Property Lifecycle Automation SaaS | *MERN Stack, Next.js, PostgreSQL*

- Developed a property management SaaS platform for tenant onboarding, invoice management, recurring billing, and rent tracking automation.
- Built responsive dashboards supporting **50+ monthly property transactions**.
- Implemented secure backend APIs and optimized database operations for improved query efficiency.